

SWEET CUPCAKE SHOP SYSTEM

USER MANUAL

Version: 1.0

Date: January 27, 2026

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Course: CSE4006 - Object Oriented Programming

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1. INTRODUCTION

The Sweet Cupcake Shop Management System is A desktop application designed to automate and Streamline Cupcake Inventory management for bakery staff. The system provides role - based access for Cashiers and Managers, enabling efficient product management and user administration.

Key Features:

- User Authentication with role-based access
- Cupcake inventory management
- Category-based search functionality
- User account creation (Manager only)
- File-based data persistence

2. SYSTEM REQUIREMENTS

2.1 Hardware Requirements:

- Processor : Intel Core i3 or equivalent
- RAM : 4GB minimum
- Storage: 100MB free Space
- Display : 1024x768 resolution minimum

2.1 Software Requirements

- Operating System: Windows 7/8/10/11, macOS, or Linux
- Java Runtime Environment (JRE) 8 or higher
- Java Development Kit (JDK) 8 or higher (for development)

3. INSTALLATION GUIDE

Step 1: Ensure Java Is Installed On your System

- Open Command Prompt/Terminal
- Type: `java -version`
- If Java is not installed, download from java.com (oracle)

Step 2: Extract the Application Folder to your computer

Step 3: Navigate to the Proper directory

Step 4: Run the Application

- Double-click `SweetCupcakeShop.jar` (if compiled)
- OR run from IDE (NetBeans/Eclipse)

4. GETTING STARTED

4.1 First Launch

When you first launch the application:

1. The system automatically creates a data folder
2. Default user accounts are created
3. Sample cupcakes are loaded for testing

4.2 Default Login Credentials

Manager Account:

- Username: admin
- Password: admin123

Cashier Account:

- Username: cashier
- Password: cashier123

4.3 Login Screen



Sweet Cupcake Shop - Login

Sweet Cupcake Shop

Login to Continue

Username:

Password:

Login **Clear**

Default: admin/admin123 or cashier/cashier123

Steps to Login:

1. Enter your username in the "Username" field
2. Enter your password in the "Password" field
3. Click the **"Login"** button
4. Or press **Enter** on your keyboard

Login Validation:

- Both username and password are required
- Credentials are case-sensitive
- Invalid credentials will show an error message

5.USER ROLES

5.1 Cashier

Permissions:

- View all cupcakes
- Add new cupcakes
- Search cupcakes by category
- Logout

5.2 Manager

Permissions:

- All Cashier permissions Plus+
- Create new cashier accounts
- View all system users

6. CASHIER FUCTIONS

6.1 Cashier Dashboard

Sweet Cupcake Shop - Cashier Dashboard

Welcome, cashier!

Role: Cashier

Logout

View All Cupcakes

Add New Cupcake

Search by Category

ID	Name	Flavor	Category	Price (\$)	Stock
CUP1769499415252	Chocolate Delight	Rich Chocolate	CLASSIC FLAVORS	3.50	50
CUP1769499415253	Vanilla Dream	Vanilla Bean	CLASSIC FLAVORS	3.00	40
CUP1769499415253	Red Velvet Supreme	Red Velvet	RED VELVET	4.00	30
CUP1769499415253	Strawberry Bliss	Fresh Strawberry	FRUITY DELIGHTS	3.75	35
CUP1769499415253	Lemon Burst	Zesty Lemon	FRUITY DELIGHTS	3.50	25
CUP1769499415253	Pumpkin Spice	Autumn Spice	SEASONAL SPECIALS	4.25	20
CUP1769499415253	Birthday Surprise	Confetti Vanilla	CUSTOM ORDERS	5.00	15
CUP1769499415253	Almond Joy	Almond	GLUTEN FREE	4.50	20
CUP1769499415253	Mint Chocolate Chip	Mint & Chocolate	SEASONAL SPECIALS	3.75	25
CUP1769499415253	Caramel Crunch	Salted Caramel	CLASSIC FLAVORS	4.00	30

After logging in as a Cashier, you'll see:

- Welcome message with your username
- Your role (Cashier)
- Four main buttons:
 -  View All Cupcakes
 -  Add New Cupcake
 -  Search by Category
 - Logout button (top right)

6.2 View All Cupcakes

Purpose: Display all cupcakes in the inventory

Steps:

1. Click "  **View All Cupcakes**" button
2. A table will populate with all cupcake data
3. Success message shows number of cupcakes loaded

Table Columns:

- ID - Unique identifier
- Name - Cupcake name
- Flavor - Flavor description
- Category - Product category
- Price (\$) - Unit price
- Stock - Available quantity

ID	Name	Flavor	Category	Price (\$)	Stock
CUP1769498924080	Chocolate Delight	Rich Chocolate	CLASSIC FLAVORS	3.50	50
CUP1769498924081	Vanilla Dream	Vanilla Bean	CLASSIC FLAVORS	3.00	40
CUP1769498924081	Red Velvet Supreme	Red Velvet	RED VELVET	4.00	30
CUP1769498924081	Strawberry Bliss	Fresh Strawberry	FRUITY DELIGHTS	3.75	35
CUP1769498924081	Lemon Burst	Zesty Lemon	FRUITY DELIGHTS	3.50	25
CUP1769498924081	Pumpkin Spice	Autumn Spice	SEASONAL SPECIALS	4.25	20
CUP1769498924081	Birthday Surprise	Confetti Vanilla	CUSTOM ORDERS	5.00	15
CUP1769498924081	Almond Joy	Almond	GLUTEN FREE	4.50	20
CUP1769498924081	Mint Chocolate Chip	Mint & Chocolate	SEASONAL SPECIALS	3.75	25
CUP1769498924081	Caramel Crunch	Salted Caramel	CLASSIC FLAVORS	4.00	30

6.3 Add new Cupcake

Purpose: Add a new cupcake to inventory

Steps:

1. Click " Add New Cupcake" button
2. A dialog window will appear
3. Fill in the required fields:
 - **Cupcake Name:** Enter the product name
 - **Flavor:** Describe the flavor
 - **Category:** Select from dropdown menu:
 - CLASSIC FLAVORS
 - RED VELVET
 - FRUITY DELIGHTS
 - SEASONAL SPECIALS
 - CUSTOM ORDERS
 - GLUTEN FREE
 - **Price (\$):** Enter price (must be positive number)
 - **Stock Quantity:** Enter stock count (must be non-negative)
 - **Description:** Optional details (multi-line)
4. Click " Save" to add the cupcake
5. Click " Clear" to reset all fields
6. Click " Cancel" to close without saving

Validation:

- All fields except Description are required
- Price must be a valid positive number
- Stock must be a valid non-negative integer
- System will show error messages for invalid input

Success:

- Confirmation dialog displays cupcake details
- Cupcake is saved to file
- Form clears for next entry

Add New Cupcake

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Add New Cupcake

Cupcake Name:

Flavor:

Category:

CLASSIC FLAVORS

▼

Price (\$):

Stock Quantity:

Description:

Save

Clear

❑ Cancel

6.4 Search Cupcakes by Category

Search Cupcakes by Category

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Search Cupcakes

Select Category

Category:

CLASSIC FLAVORS

▼

Search

Clear

Search results will appear here

ID	Name	Flavor	Category	Price (\$)	Stock
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Close

Purpose: Find cupcakes in a specific category

Steps:

1. Click " **Search by Category**" button
2. Search dialog will appear
3. Select a category from the dropdown
4. Click " **Search**" button
5. Results appear in the table below
6. Result count is displayed

Features:

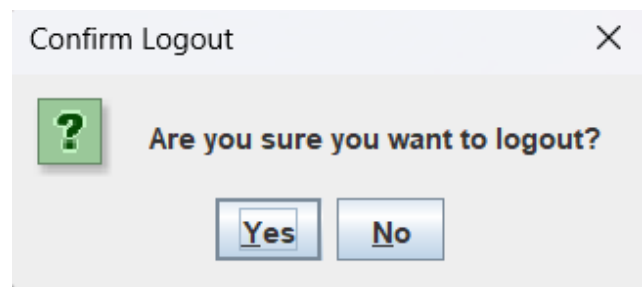
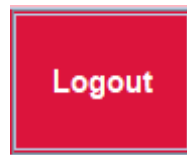
- Shows all cupcakes in selected category
- Displays count of results found
- Click " **Clear**" to reset search
- Click " **Close**" to exit

No Results:

- If no cupcakes found, message displays
- Try selecting a different category

6.5 Logout

Purpose: Exit the system securely



Steps:

1. Click "**Logout**" button (top right corner)
2. Confirmation dialog appears
3. Click "**Yes**" to logout
4. You'll return to the Login screen

7.MANAGER FUNCTIONS

7.1 Manager Dashboard

Sweet Cupcake Shop - Manager Dashboard

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Welcome, admin!
Role: Manager (Full Access)

Logout

View All Cupcakes

Add New Cupcake

Search by Category

Manager Functions

Create Cashier Account

View All Users

ID	Name	Flavor	Category	Price (\$)	Stock
CUP1769498924080	Chocolate Delight	Rich Chocolate	CLASSIC FLAVORS	3.50	50
CUP1769498924081	Vanilla Dream	Vanilla Bean	CLASSIC FLAVORS	3.00	40
CUP1769498924081	Red Velvet Supreme	Red Velvet	RED VELVET	4.00	30
CUP1769498924081	Strawberry Bliss	Fresh Strawberry	FRUITY DELIGHTS	3.75	35
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CUP1769498924081	Mint Chocolate Chip	Mint & Chocolate	SEASONAL SPECIALS	3.75	25
CUP1769498924081	Caramel Crunch	Salted Caramel	CLASSIC FLAVORS	4.00	30

After logging in as Manager, you'll see:

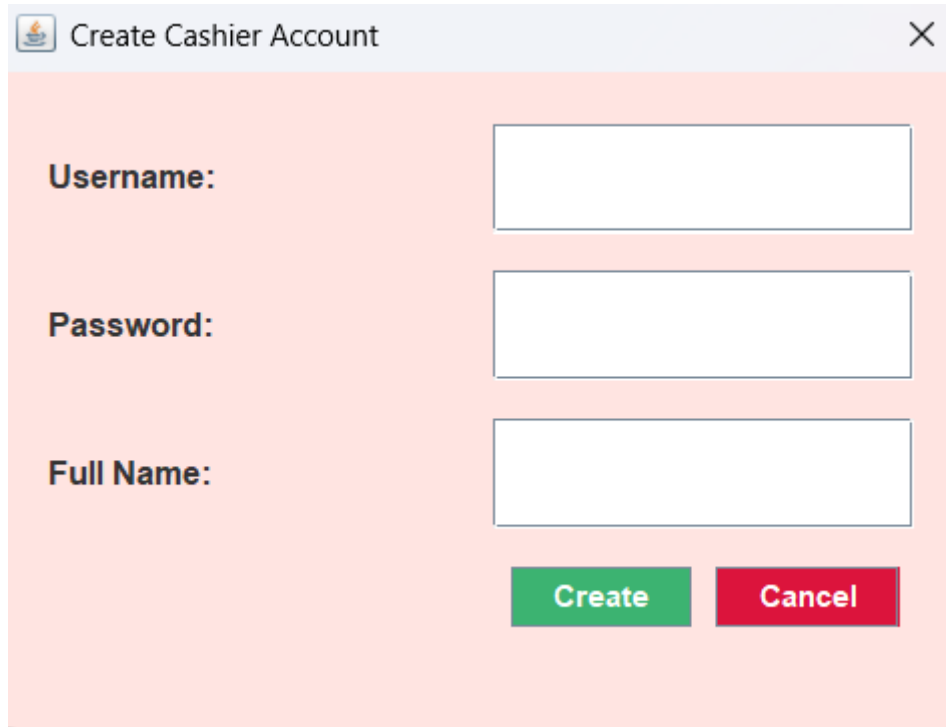
- All Cashier features PLUS
- **Manager Functions** section with two additional buttons:
 1.  Create Cashier Account
 2.  View All Users

7.2 Create Cashier Account

Purpose: Add new cashier users to the system

Steps:

1. Click "**+ Create Cashier Account**" button
2. Dialog window appears



The screenshot shows a dialog window titled "Create Cashier Account". It features three input fields for "Username:", "Password:", and "Full Name:". At the bottom, there are two buttons: "Create" (green) and "Cancel" (red).

3. Enter the following:
 - **Username:** Unique username (will be checked)
 - **Password:** Secure password
 - **Full Name:** Cashier's full name
4. Click "**Create**" to save
5. Click "**Cancel**" to abort

Validation:

- All fields are required
- Username must be unique
- System checks for duplicate usernames
- Success message displays new account details

Important Notes:

- Write down credentials immediately
- Usernames cannot be changed later
- New cashiers can login immediately

7.3 View All Users

Purpose: Display all system users

Steps:

1. Click "  View All Users" button
2. Dialog with user table appears

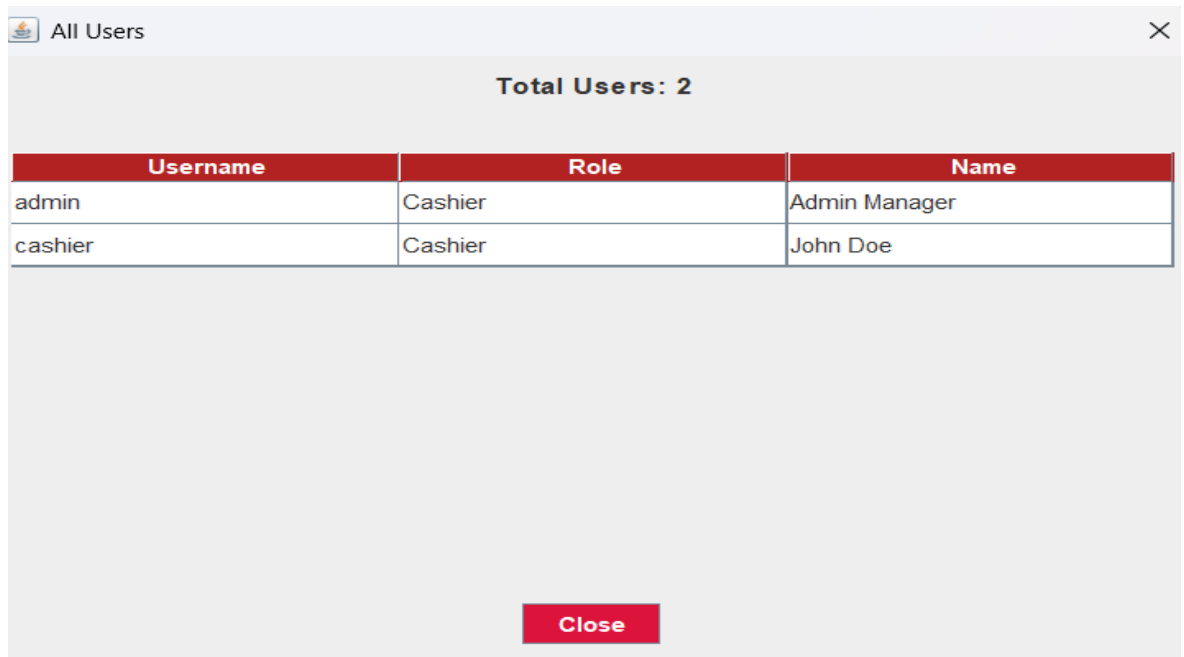


Table Shows:

- Username
- Role (Manager/Cashier)
- Name

Features:

- Total user count displayed
- Read-only table (no editing)
- Click "Close" to exit

8. TROUBLESHOOTING

Common Issues and Solutions

Problem: Cannot login **Solution:**

- Check username/password spelling
- Ensure Caps Lock is off
- Try default credentials
- Check if user account exists

Problem: Cupcakes not displaying **Solution:**

- Click "View All Cupcakes" to refresh
- Check if cupcakes.txt file exists in data folder
- Verify sample data was created on first launch

Problem: Cannot add cupcake **Solution:**

- Fill all required fields
- Ensure price is valid number
- Ensure stock is valid integer
- Check file permissions in data folder

Problem: Search returns no results **Solution:**

- Verify cupcakes exist in selected category
- Try different category
- Click "View All Cupcakes" to see all available

Problem: Application won't start **Solution:**

- Verify Java is installed (java -version)
- Check Java version (requires Java 8+)
- Run from command line to see error messages
- Check for missing library files

====END OF USER MANUAL====